SOMETHING ABOUT ANTS

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"The Subject indeed is small, but not inglorious. The Ant, as the Prince of Wisdom is pleased to inform us, is exceedingly wise. In this Light it may, without Vanity, boast of its being related to you, and therefore by right of Kindred merits your protection"

– William Gould, *"An Account of English Ants,"* 1747

ABSTRACT. Rough Draft: We explore optimal resource allocation strategies for ant colonies at varying levels of complexity and under different constraints. Gotta flesh this out better, but yeah that's basically the idea. Place abstract here. The abstract summarizes in one paragraph the main question and conclusions draw from your investigation.

1. Background

There exist an estimated ten thousand trillion ants (family Formicidae) across over ten thousand species. These organisms rival humanity in terms of biomass and geographic reach. They have adapted to thrive in subterranean depths as well as high forest canopies, in arid cold as well as tropical heat. Even as entire ecosystems collapse in the wake of an ongoing great extinction event, ants persist not only as survivors, but as ecological stabilizers in their respective habitats [HW94, OW78]. Myrmecologists have long suggested that ants' rank as dominant organisms is largely due to their eusocial nature and communication abilities [Whe10]. In these ways, ants closely resemble mankind. Our similarities do not stop there; both of our societies are characterized by similar objectives of economic growth and resource optimization, pursued through analogous systems of decentralized control. Just as humanity relies on the constructs of law and culture to organize collective effort, ant colonies achieve highly efficient labor allocation and social stabilization through complex pheromonal networks.

The use of optimal control theory to model social insect populations is notably seen established in 1976 by myrmecologists Macevicz and Oster who demonstrated a bang-bang solution for a colony's seasonal shift from an ergonomic growth phase (wherein a colony is investing in workers and growth) a reproductive phase, (wherein, prior to overwintering, a colony ceases to develop workers and instead prioritizes investment in reproducer ants for the following season) [MO76, OW78]. While this and related models have demonstrated insightful results, they treat colonies as isolated entities.

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We could find no documentation of optimal control theory applied in settings with competing colonies, though given the prevalence of warfare among ants we expect this would effect population dynamics and resource allocation in significant ways [OW78].

The purpose of this exposition is to investigate optimal resource allocation and reproduction strategies for mature ant colonies under a strict seasonal time horizon. We develop and analyze optimal control models at varying levels of complexity. First, we establish a baseline by formulating our own isolated-colony model. Second, we expand the state space to evaluate optimal resource policies between two rival colonies. By framing the rival colony's behavior as a predetermined, time-dependent strategy, we explore the optimal dynamic responses of our primary colony in a competitive environment. Ultimately, this investigation provides a theoretical framework for competitive resource allocation that extends to economic strategies and other human applications.

2. Mathematical Representation

We begin by defining the control variable u as the rate at which the queen produces new worker ants. The system is characterized by two state variables: the worker population (x_1) and the available resources (x_2). The state equations incorporate several key parameters, including the worker death rate (δ), the per-ant resource collection rate (γ), the maintenance rate required to sustain the worker population (m), and the cost associated with producing new workers (c). In extending the model to account for two competing colonies, we introduce territory as a function of the worker population size. Specifically, we define a parameter (p) representing the density of ants per unit radius from the colony, which enables estimation of inter-colony contact. Additionally, we include a competition-induced loss rate (α) to capture the effects of interactions between colonies.

To address this problem, the objective is not solely to maximize the total resources accumulated by a colony at the end of a season. Rather, the formulation accounts for the trade-off between maintaining a sufficiently large worker population during the active season to ensure colony defense and the inefficiency associated with surplus workers remaining at the end of the season. Additionally, the production of worker ants imposes a reproductive cost on the queen, which must also be incorporated into the model. We imposed this cost in response to Oster's findings regarding the queen's physiological limits for both daily and lifelong reproduction; a queen ant is generally only inseminated once, during her nuptial flight, and depends on that single store of sperm to grow and maintain colony population and growth throughout her life. In some cases this lasts decades [OW78].

These considerations lead to the following cost functional:

$$J[u] = \int_0^{t_f} \left(-x_1 + u^2\right) \, dt + x_1(t_f) - x_2(t_f).$$

The Mayer terms of this cost functional induce a transversality condition on the costate variables. In particular, we obtain

$$\frac{\partial \phi}{\partial x(t_f)} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

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2.1. **Isolated Colony.** For modeling the optimal control of ant colony worker recruitment for a single colony, the state space is two-dimensional, capturing the worker population and the total resource reserve. The state evolution equations are

$$\begin{aligned} \dot{x}_1 &= -\delta x_1 + u, \\ \dot{x}_2 &= \gamma x_1 - m x_1 - c u, \end{aligned}$$

where x_1 denotes the worker population, x_2 the resource level, $u \geq 0$ the recruitment rate, δ the worker decay rate, γ the resource gathering rate per worker, m the cost of resource maintenance and c the recruitment/reproduction cost per worker.

2.2. **Rival Colonies.** For modeling competing colonies, we introduced two additional state variables x_3 and x_4 to represent the workers and resources of the rival colony, respectively. We decided to relate the amount of direct competition between the colonies to the amount of contact between rival workers. We let these state variables evolve similarly to those of our base colony, with one caveat. Since we control (and wish to optimize) the reproduction strategy of the base colony and not that of the rival, we decided to let $S(t)$ be a predetermined reproduction strategy for the rival colony. The problem then resembles game theory, as we can experiment with different strategies for the opponent to see how that impacts our own colony's strategy.

We assumed that the distance between colony centers is a predetermined constant d, and that the amount of territory that a colony's ants cover is proportional to the number of ants in that colony. Thus, the amount of competition between colonies can be modeled a function of x_1 and x_3. Specifically we define a conflict function $W(x_1, x_3) = e^{p_1 x_1 + p_2 x_3 - d}$.

In order to account for the tax of competition on the worker ant population, we incorporated $-W(x_1, x_3)$ into our state evolution equations to represent deaths due to competition.

Our newly adjusted state evolution equations become

$$\begin{aligned} \dot{x}_1 &= -\delta_1 x_1 + u - \alpha_1 x_1 W(x_1, x_3), & \dot{x}_2 &= \gamma_1 x_1 - m_1 x_1 - c_1 u, \\ \dot{x}_3 &= -\delta_2 x_3 + s(t) - \alpha_2 x_3 W(x_1, x_3), & \dot{x}_4 &= \gamma_2 x_3 - m_2 x_3 - c_2 s(t). \end{aligned}$$

2.3. **LQR Approach.** We hoped to derive a formulation to which we could apply an LQR solution. We were drawn to this approach for the rival colony case, as in a real-world setting a base colony would not know the exact state values of its rival and this formulation would have allowed us to implement a more realistic LQG control, using the Kalman Filter to estimate the rival colony's state values. Our rival colonies model first violated the conditions of LQR in its state evolution equations. It is straightforward to see that these are highly are nonlinear. We aimed to combat this by linearizing these locally. More critically, however, our cost functional violated the quadratic conditions of LQR. We could find no way to reformulate this without forfeiting what we supposed to be the true objectives of our base colony. Hence, we abandoned this approach.

3. Solution

3.1. **Isolated Colony.**

The Hamiltonian corresponding to the cost functional and our isolated colony state evolution equations is constructed as

$$H = \lambda_1(-\delta x_1 + u) + \lambda_2(\gamma x_1 - m x_1 - cu) - x_1 + u^2,$$

where the running cost $L = -x_1 + u^2$ reflects the objective of maximizing the worker population while penalizing costly recruitment. Applying the costate equations $\dot{\lambda} = -\partial H/\partial x$ yields

$$\begin{aligned}\dot{\lambda}_1 &= \lambda_1\delta - \lambda_2\gamma + 1,\\ \dot{\lambda}_2 &= 0\end{aligned}$$

With transversality conditions

$$\lambda(t_f) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Differentiating the Hamiltonian with respect to u and applying Pontryagin's Maximum Principle allow us to find the optimal control of our system as a function of the costates:

$$\begin{aligned}\frac{\partial H}{\partial u} &= \lambda_1 - c\lambda_2 + 2u = 0\\ u(t) &= \frac{c\lambda_2 - \lambda_1}{2}.\end{aligned}$$

The condition $\dot{\lambda}_2 = 0$ implies that λ_2 is constant, allowing for a more straightforward derivation of the solution. Applying the boundary conditions yields $\lambda_2 = C = -1$. With this result, λ_1 can be computed numerically using `solve_ivp`. Substituting these costate values back into the control expression u and the state equations then yields the complete solution.

3.2. **Rival Colonies.**

The Hamiltonian corresponding to our cost functional with our rival colony model's state evolution equations was

$$H = \lambda^T \begin{bmatrix} -\delta_1 x_1 + u - \alpha_1 x_1 W(x_1, x_3) \\ \gamma_1 x_1 - m_1 x_1 - c_1 u \\ -\delta_2 x_2 + s(t) - \alpha_2 x_3 W(x_1, x_3) \\ \gamma_2 x_3 - m_2 x_3 - c_2 s(t) \end{bmatrix} + u^2 - x_1$$

Applying Hamilton's equations then gave us our costate equations

$$\dot{\lambda}_1 = -(-\delta_1\lambda_1 - \alpha_1\lambda_1(W + p_1x_1W) + \lambda_2 + \gamma_1 - \lambda_2 m_1 + 1) \qquad \dot{\lambda}_2 = 0$$
$$\dot{\lambda}_3 = -(-\delta_2\lambda_3 - \alpha_2\lambda_3(W + p_2x_3W) + \lambda_4\gamma_2 - \lambda_4 m_2) \qquad \dot{\lambda}_4 = 0$$

We solved this problem directly with `solve_bvp` to both the state costate evolution equations. This approach was very similar to the process outlined in Chapter 19 of the book. We can also derive that

$$\frac{\partial H}{\partial u} = \lambda_1 - c_1\lambda_2 + 2u = 0$$

The equation for u looks almost identical to the isolated colony because the state equations of the rival colony are not in terms of u.

4. Interpretation

4.1. **Isolated Colony.** We note that the system appears to exhibit a natural carrying capacity even without explicitly imposing one in the model formulation. The resulting behavior aligns with prior expectations of the control (representing the introduction of new worker ants). We see that reproduction increases towards an equilibrium state, where it remains throughout the majority of the season until tapering off as the colony prepares for overwintering. The model produces insights into how to maximize output while avoiding excessive expansion of the worker ants. In particular, the optimal strategy involves increasing the number of workers to a target level, maintaining that level for the majority of the season, and then gradually tapering off new worker ant production as the end of the season approaches. This framework can be generalized to other settings. For example, private equity firms typically hold portfolio companies for a period of three to five years and seek to maximize growth without overexpanding employees. By adjusting model parameters—such as reducing the attrition rate to reflect lower employee turnover and allowing for a free terminal condition—the same optimal control structure can be applied. Under these modifications, we would expect a similar qualitative result, where hiring occurs at a relatively steady rate before tapering near the end of the investment horizon.

4.2. **Rival Colony.** We started by testing the rival strategy of $S(t) = 0.8$. Setting a constant produced nearly identical solutions as the rival emulating our semi bang-bang solution from the isolated colony.

Another rival strategy scenario we tested was to represent the situation that the rival colony got off to a slow start. The slow start was followed by an ramp up for the rest of the season to regain lost time. Thus, we let $s(t) = (\frac{t}{2})^2$.

We can see that $u^\star$ changes to take advantage of the rival's slow start. New worker ants are created at a much higher rate at the start of the season, with a drop off in new worker ants as the season goes on and the rival colony

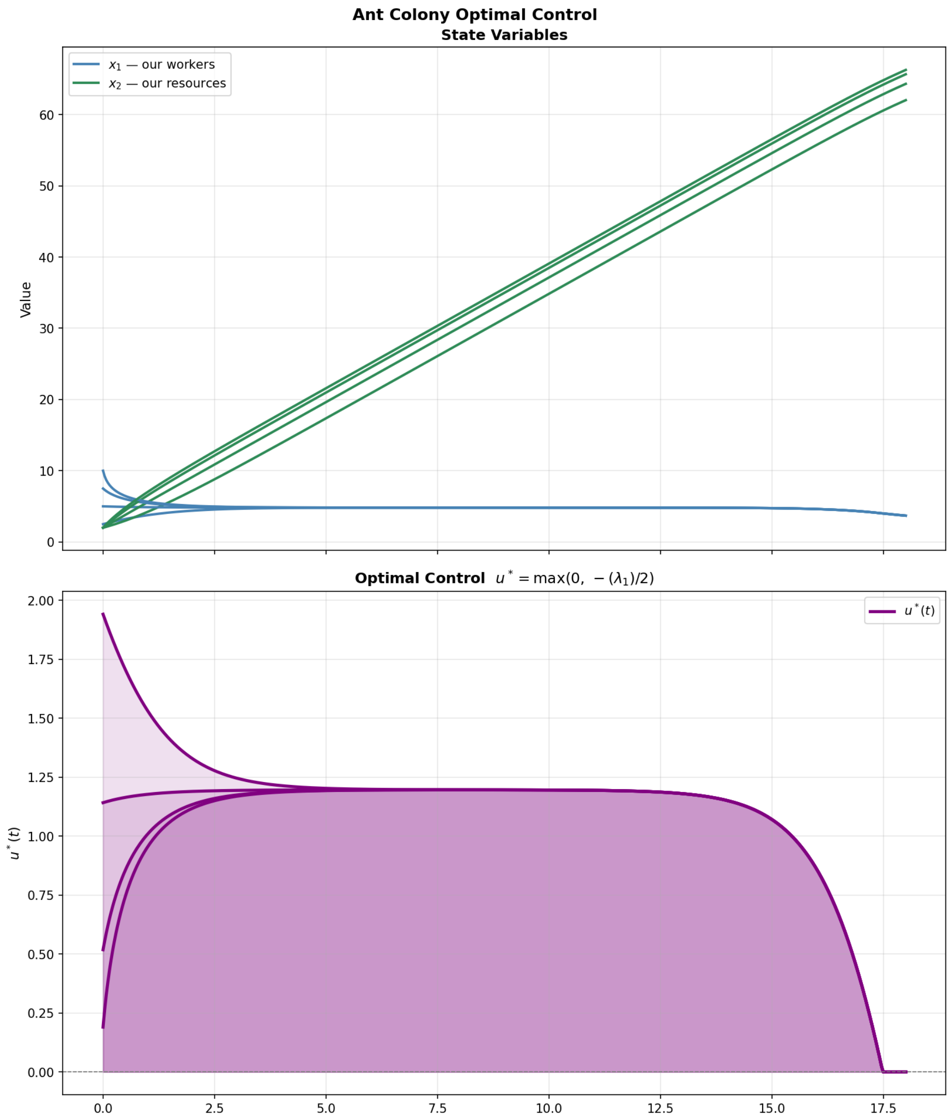

FIGURE 1. Optimal control $u(t)$ for the isolated-colony model. The recruitment rate remains high for most of the season and drops rapidly near the terminal time.

gains strength. Also note that with all other parameters constant, we gain more resources as a consequence of less initial competition.

Finally, we tested a rival strategy that involved an exponential decay. The decay models that a colony is declining over the season instead of maintaining a constant rate.

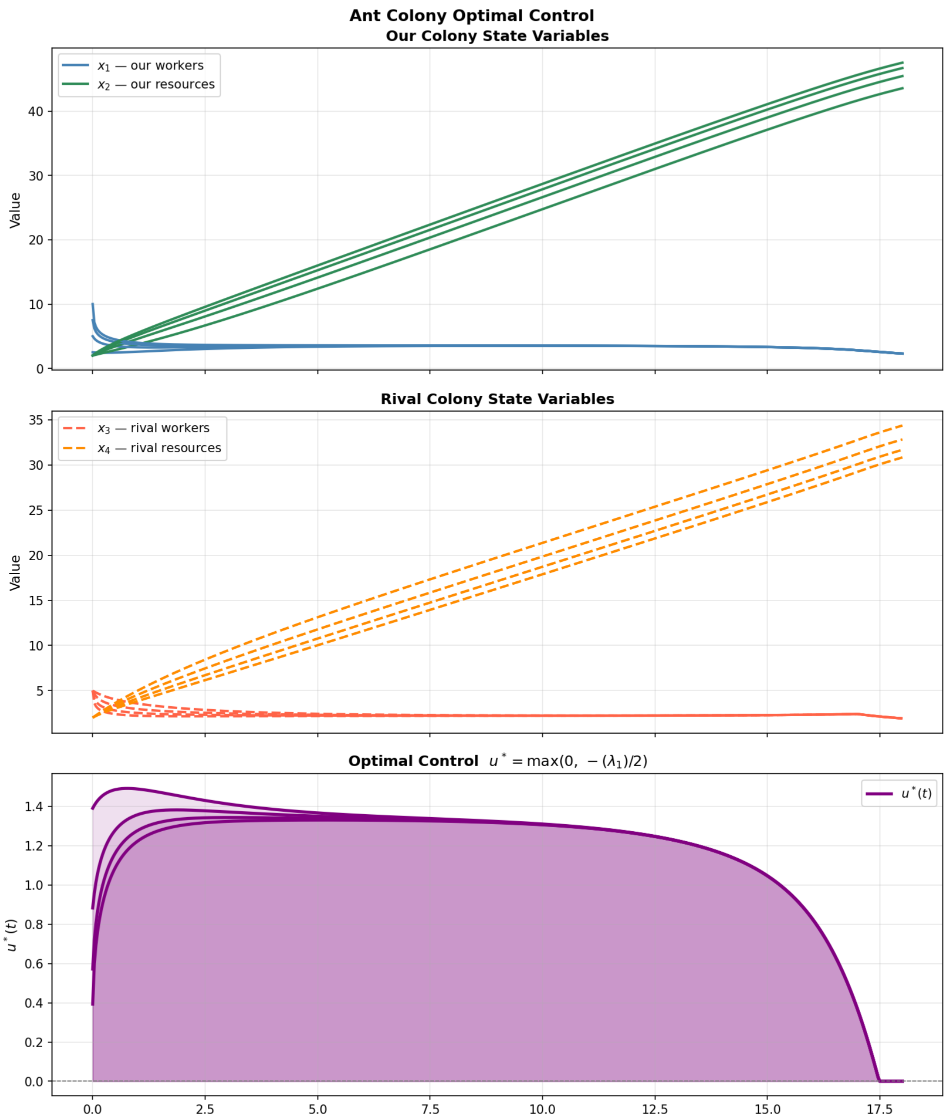

FIGURE 2. Solution with rival colony implementing bang-bang strategy emulating the optimal strategy from the isolated colony. Initial conditions for our colony's ants were 2.5, 5, 7.5, and 10 (in thousands of workers). All other parameters assumed equal between two colonies

"We give Dr. Jarvis, and other instructors at BYU teaching ACME classes permission to share our project as an example of a good project in future classes they teach."

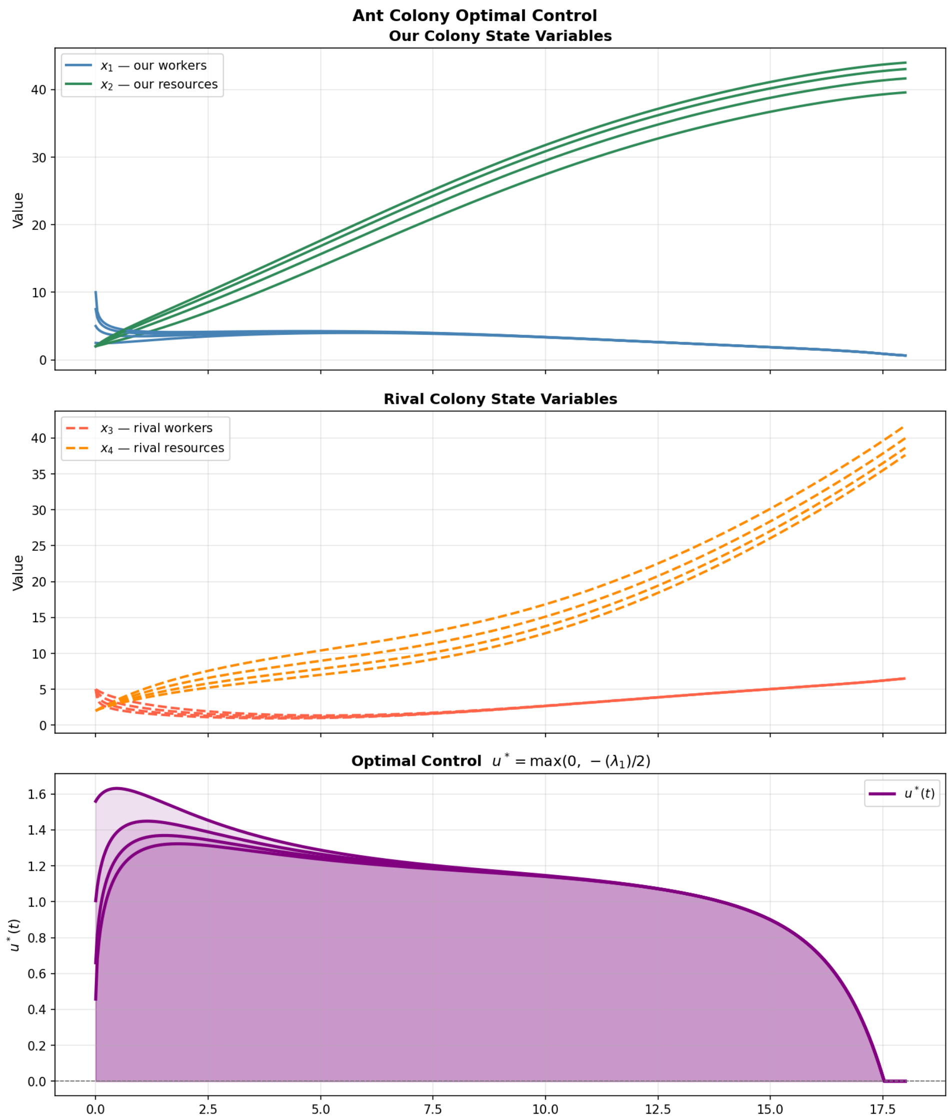

FIGURE 3. Solution with rival colony implementing ramp-up strategy over time. Initial conditions for our colony's ants were 2.5, 5, 7.5, and 10 (in thousands of workers). All other parameters assumed equal between two colonies

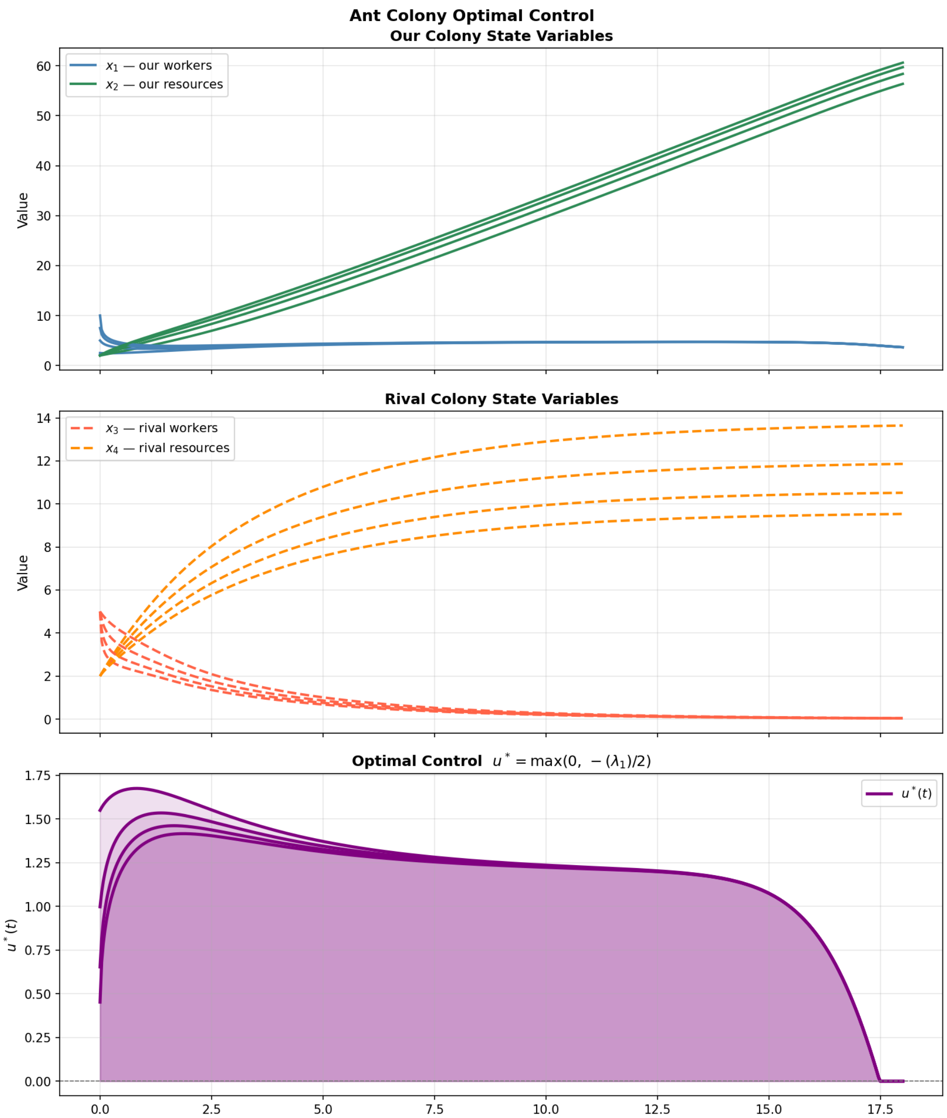

FIGURE 4. Solution with rival colony implementing decay strategy over time. Initial conditions for our colony's ants were 2.5, 5, 7.5, and 10 (in thousands of workers). All other parameters assumed equal between two colonies

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